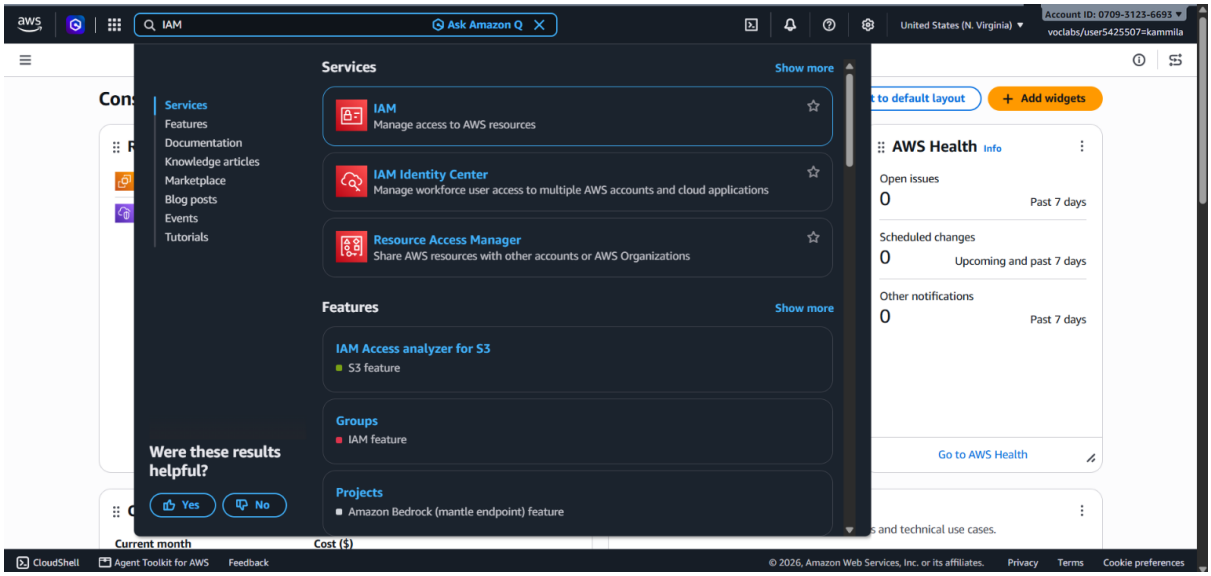
Practical 6

6. Introduction to AWS Identity and Access Management (IAM)

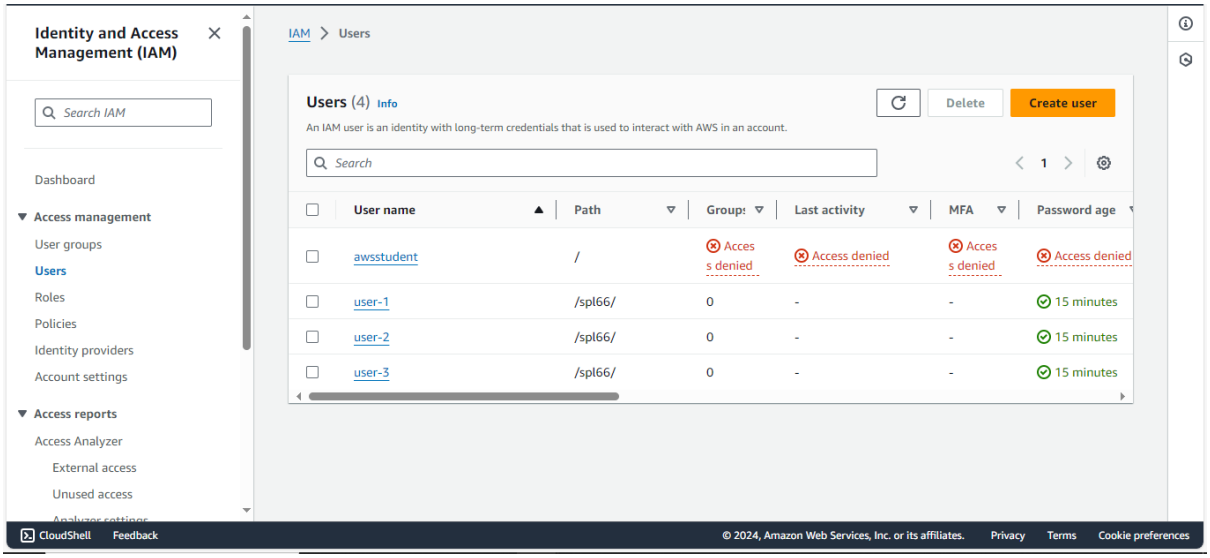
Step 1: Explore the Users and Groups

In this task, you will explore the Users and Groups that have already been created for you in IAM.

- In the search box to the right of **Services**, search for and choose **IAM** to open the IAM console.



- In the navigation pane on the left, choose **Users**.

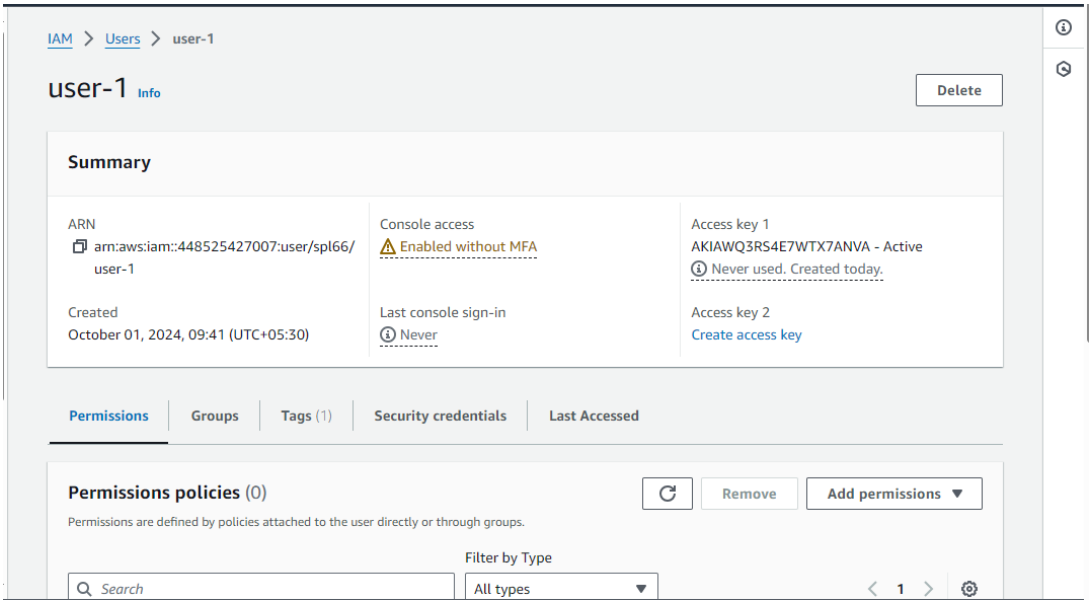


The following IAM Users have been created for you:

- user-1
- user-2
- user-3

- Choose the **user-1** link.

This will bring to a summary page for user-1. The **Permissions** tab will be displayed.



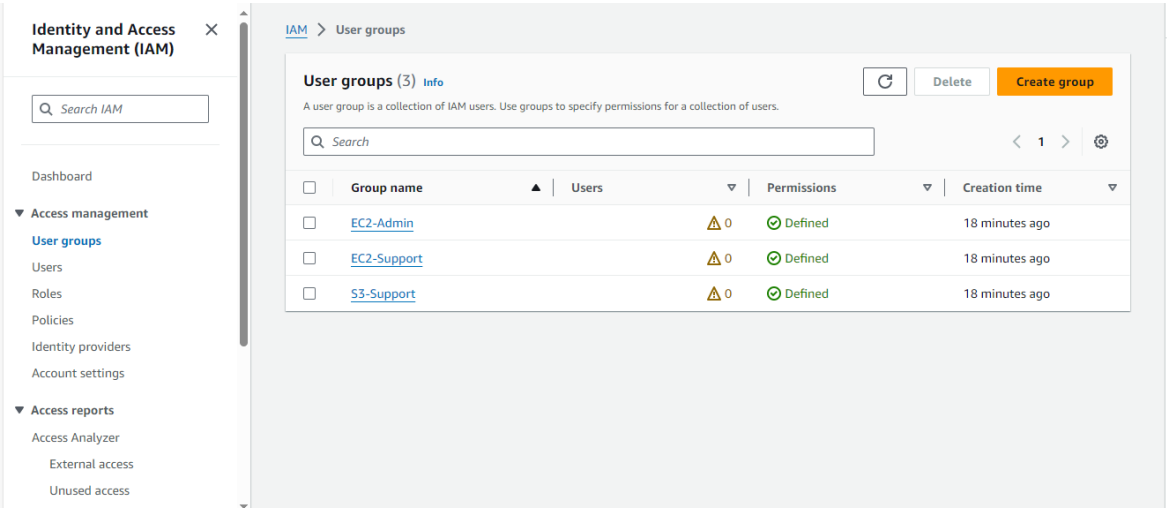
- Notice that user-1 does not have any permissions.
- Choose the **Groups** tab.

user-1 also is not a member of any groups.

- Choose the **Security credentials** tab.

user-1 is assigned a **Console password**.

- In the navigation pane on the left, choose **User groups**.

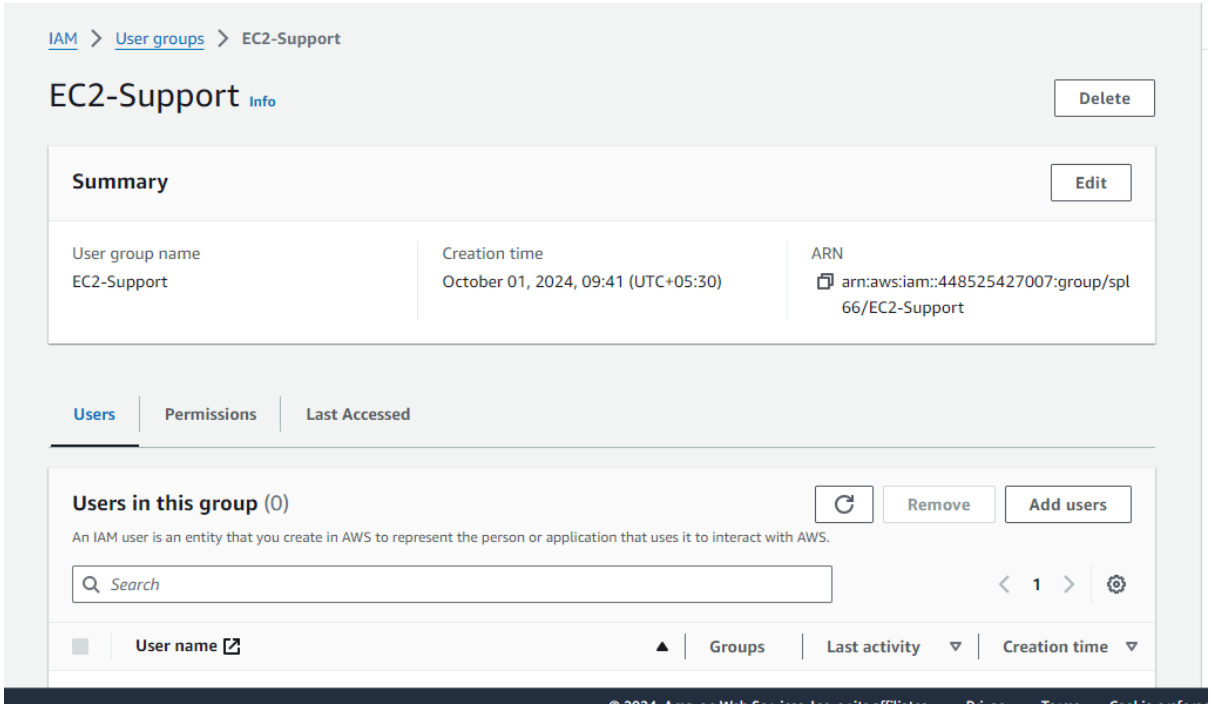


The following groups have already been created for you:

- EC2-Admin
- EC2-Support
- S3-Support

- Choose the **EC2-Support** group link.

This will bring you to the summary page for the **EC2-Support** group.

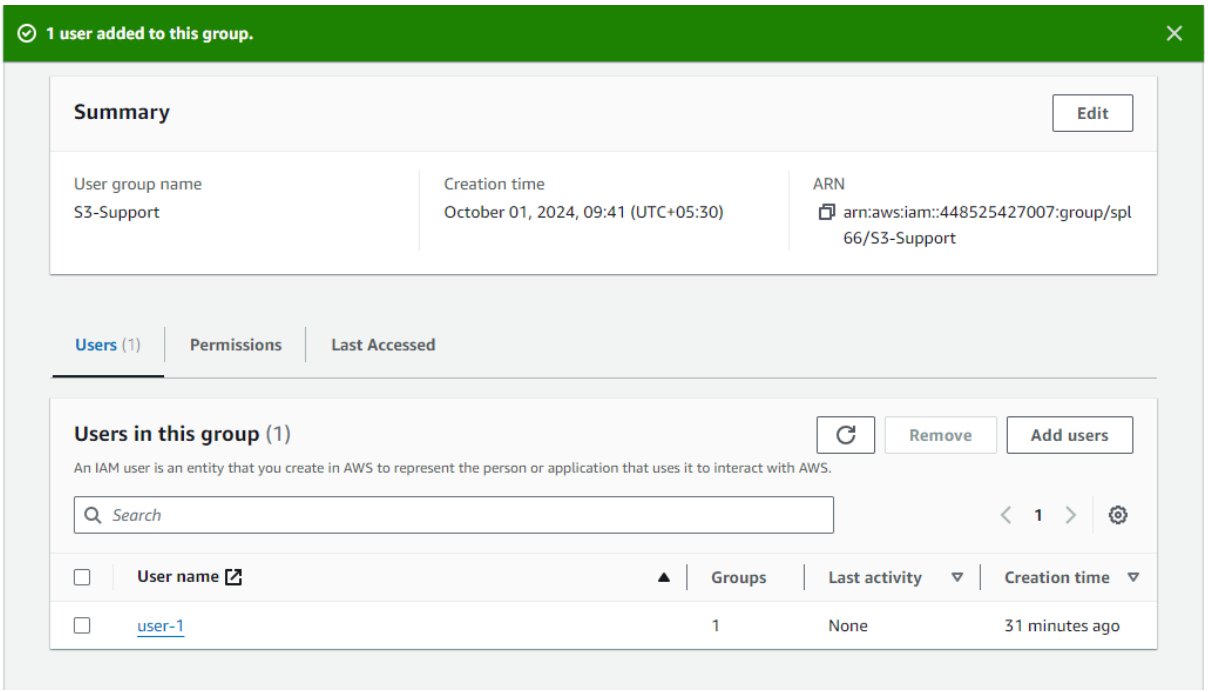


Step 2:
Add Users to Groups
You have recently hired **user-1** into a role where they will provide support for Amazon S3. You will add them to the **S3-Support** group so that they inherit the necessary permissions via the attached *AmazonS3ReadOnlyAccess* policy.

Add user-1 to the S3-Support Group

- In the left navigation pane, choose **User groups**.
- Choose the **S3-Support** group link.
- Choose the **Users** tab.
- In the **Users** tab, choose **Add users**.
- In the **Add Users to S3-Support** window, configure the following:
  - Select **user-1**.
  - At the bottom of the screen, choose **Add users**.

In the **Users** tab you will see that user-1 has been added to the group.

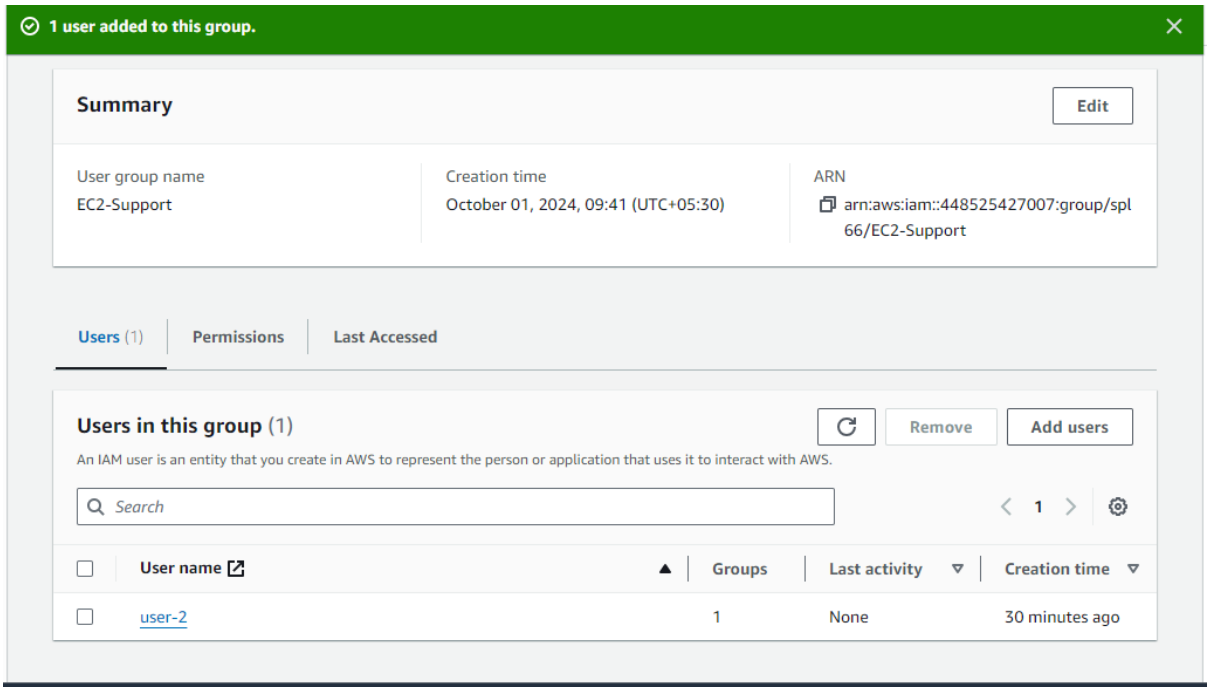


Add user-2 to the EC2-Support Group

You have hired **user-2** into a role where they will provide support for Amazon EC2.

- Using similar steps to the ones above, add **user-2** to the **EC2-Support** group.

user-2 should now be part of the **EC2-Support** group.



Add user-3 to the EC2-Admin Group

You have hired **user-3** as your Amazon EC2 administrator, who manage your EC2 instances.

- Using similar steps to the ones above, add **user-3** to the **EC2-Admin** group.

user-3 should now be part of the **EC2-Admin** group.

- In the navigation pane on the left, choose **User groups**.

Each Group should now have a **1** in the Users column, indicating the number of Users in each Group.

Step 3: Sign-In and Test Users

In this task, you will test the permissions of each IAM User.

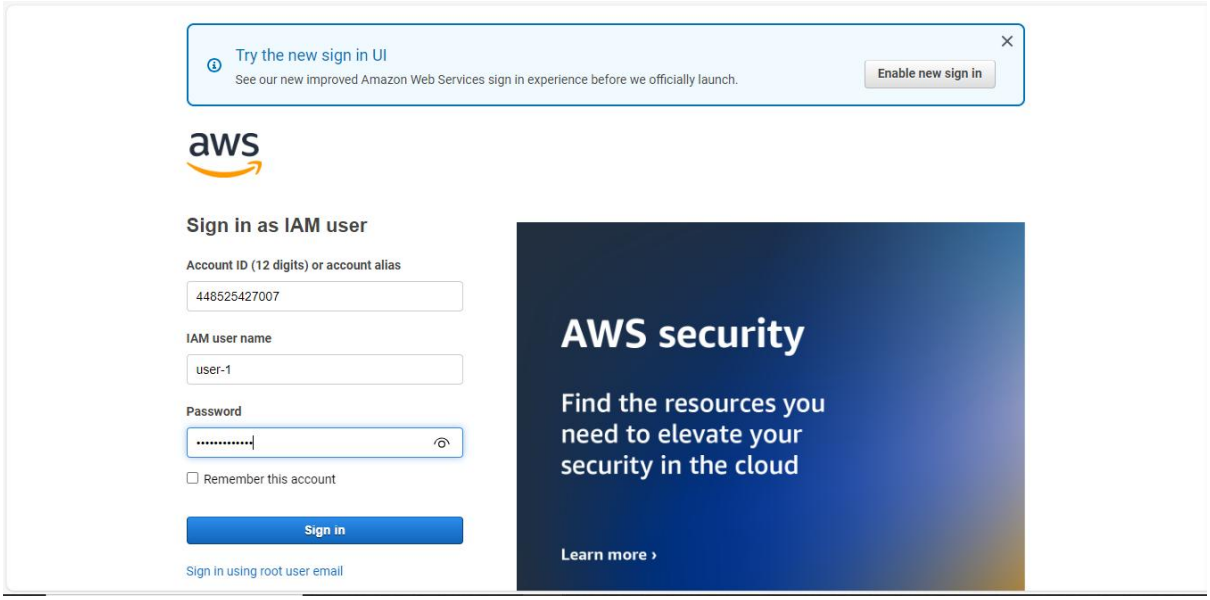
- In the navigation pane on the left, choose **Dashboard**.

A **Sign-in URL for IAM users in this account** link is displayed on the right. It will look similar to: *https://123456789012.signin.aws.amazon.com/console*
This link can be used to sign-in to the AWS Account you are currently using.

- Copy the **Sign-in URL for IAM users in this account** to a text editor.
- Open a private (Incognito) window.
- Paste the **IAM users sign-in** link into the address bar of your private browser session and press **Enter**.

Next, you will sign-in as **user-1**, who has been hired as your Amazon S3 storage support staff.

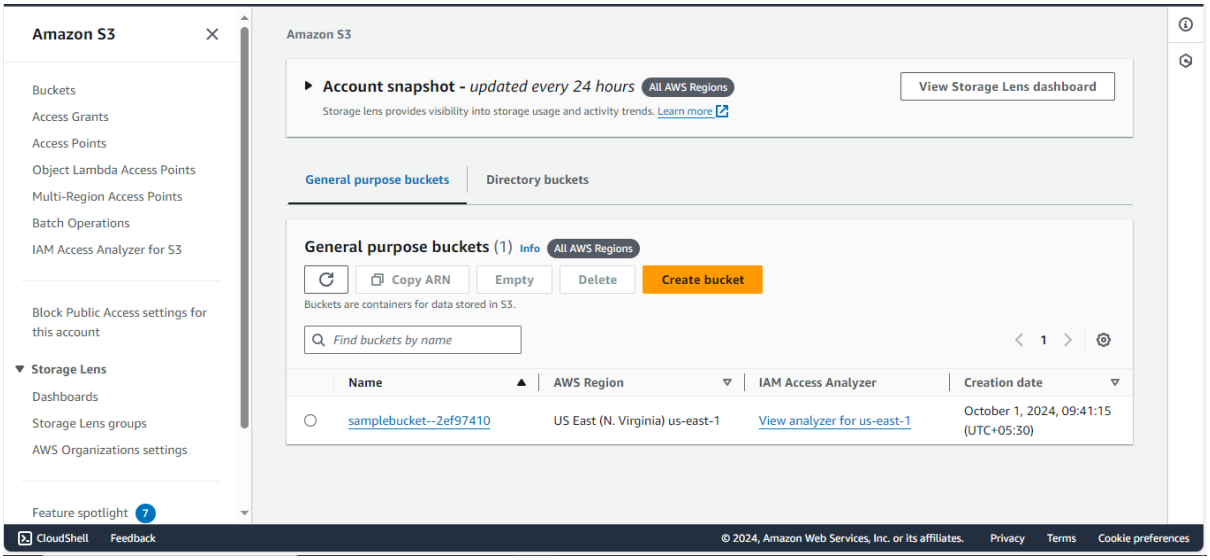
- Sign-in with:
  - **IAM user name:** user-1
  - **Password:** Lab-Password1



- In the search box to the right of **Services**, search for and choose **S3** to open the S3 console.

- Choose the name of the bucket that exists in the account and browse the contents.

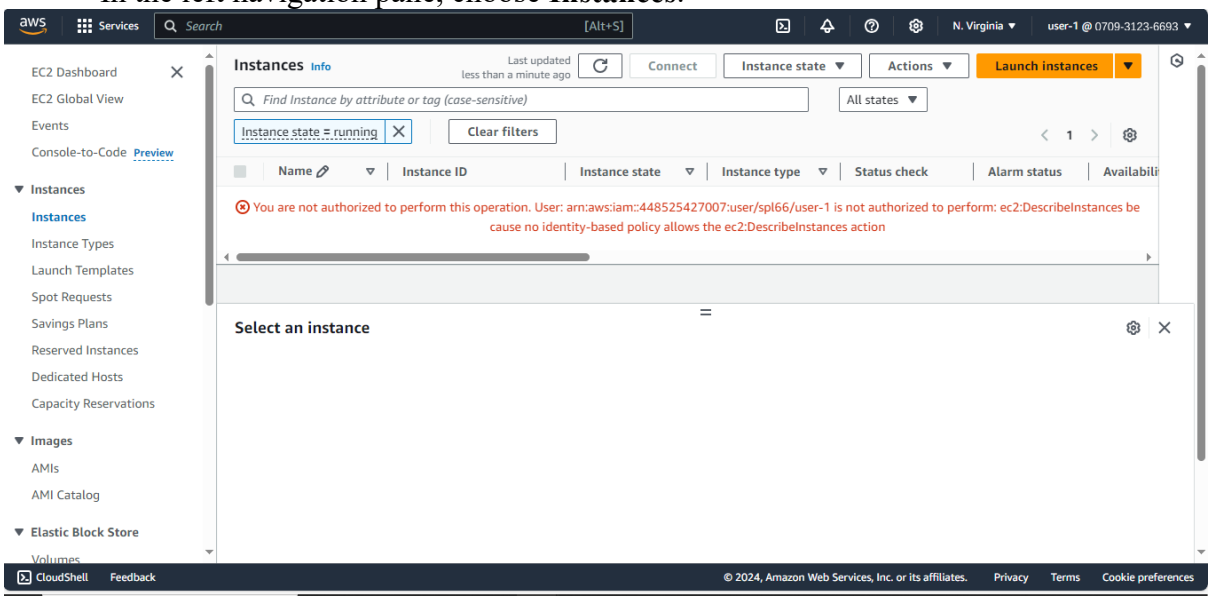
Since your user is part of the **S3-Support** Group in IAM, they have permission to view a list of Amazon S3 buckets and the contents.



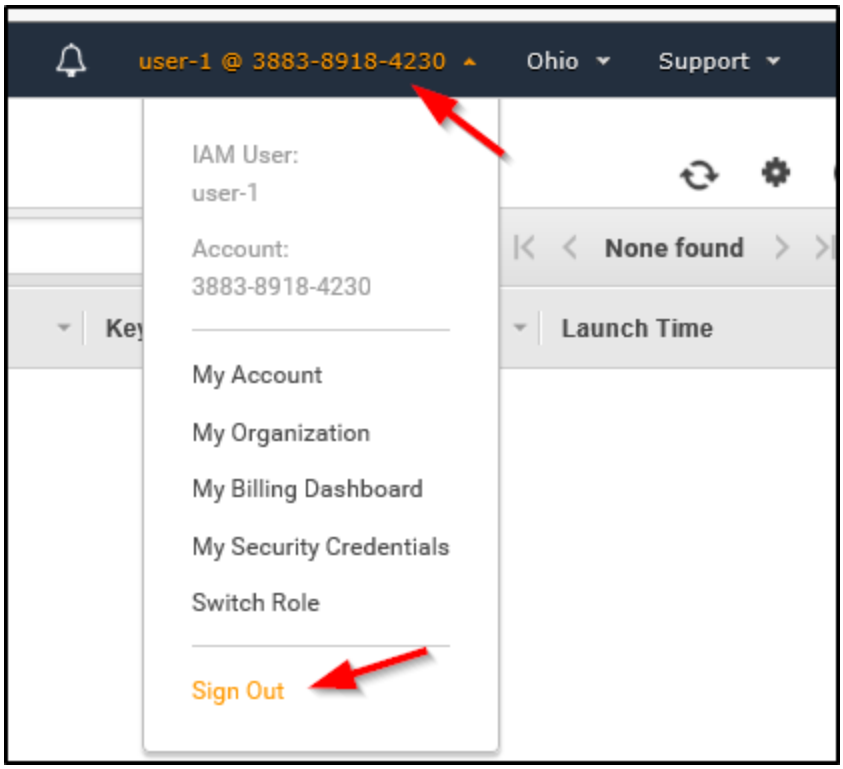
Note: The bucket does not contain any objects.
Now, test whether they have access to Amazon EC2.

- In the search box to the right of **Services**, search for and choose **EC2** to open the EC2 console.

- In the left navigation pane, choose **Instances**.



You cannot see any instances. Instead, you see a message that states *You are not authorized to perform this operation*. This is because this user has not been granted any permissions to access Amazon EC2.



- Paste the **IAM users sign-in** link into your private browser tab's address bar and press **Enter**.

Note: This link should be in your text editor.

- Sign-in with:
  - **IAM user name:** user-2
  - **Password:** Lab-Password2

- In the search box to the right of **Services**, search for and choose **EC2** to open the EC2 console.

- In the navigation pane on the left, choose **Instances**.

You are now able to see an Amazon EC2 instance because you have Read Only permissions. However, you will not be able to make any changes to Amazon EC2 resources.
If you cannot see an Amazon EC2 instance, then your Region may be incorrect. In the top-right of the screen, pull-down the Region menu and select the region that you noted at the start of the lab (for example, **N. Virginia**).

  - Select the instance named *LabHost*.

- In the **Instance state** menu above, select **Stop instance**.

- In the **Stop Instance** window, select **Stop**.

You will receive an error stating *You are not authorized to perform this operation*. This demonstrates that the policy only allows you to view information, without making changes.

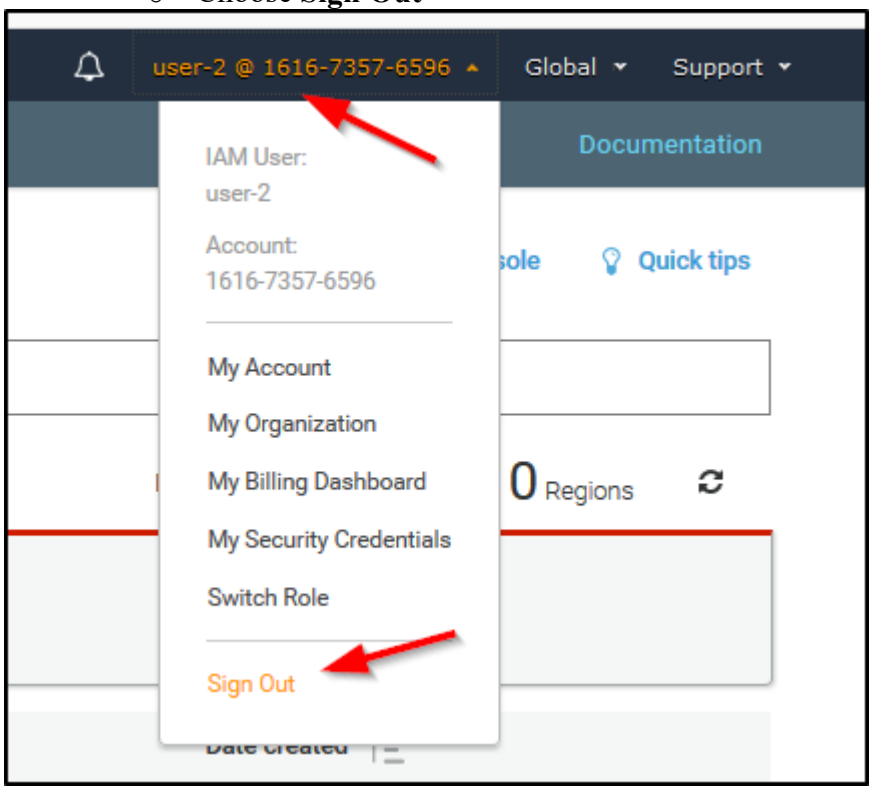
- Choose the X to close the *Failed to stop the instance* message.

Next, check if user-2 can access Amazon S3.

- In the search box to the right of **Services**, search for and choose **S3** to open the S3 console.

You will see the message **You don't have permissions to list buckets** because user-2 does not have permission to access Amazon S3.
You will now sign-in as **user-3**, who has been hired as your Amazon EC2 administrator.

- Sign user-2 out of the **AWS Management Console** by completing the following actions:
  - At the top of the screen, choose **user-2**
  - Choose **Sign Out**



- Close your private browser window.